Chapter 5

MOS Field-Effect Transistors (MOSFETs)

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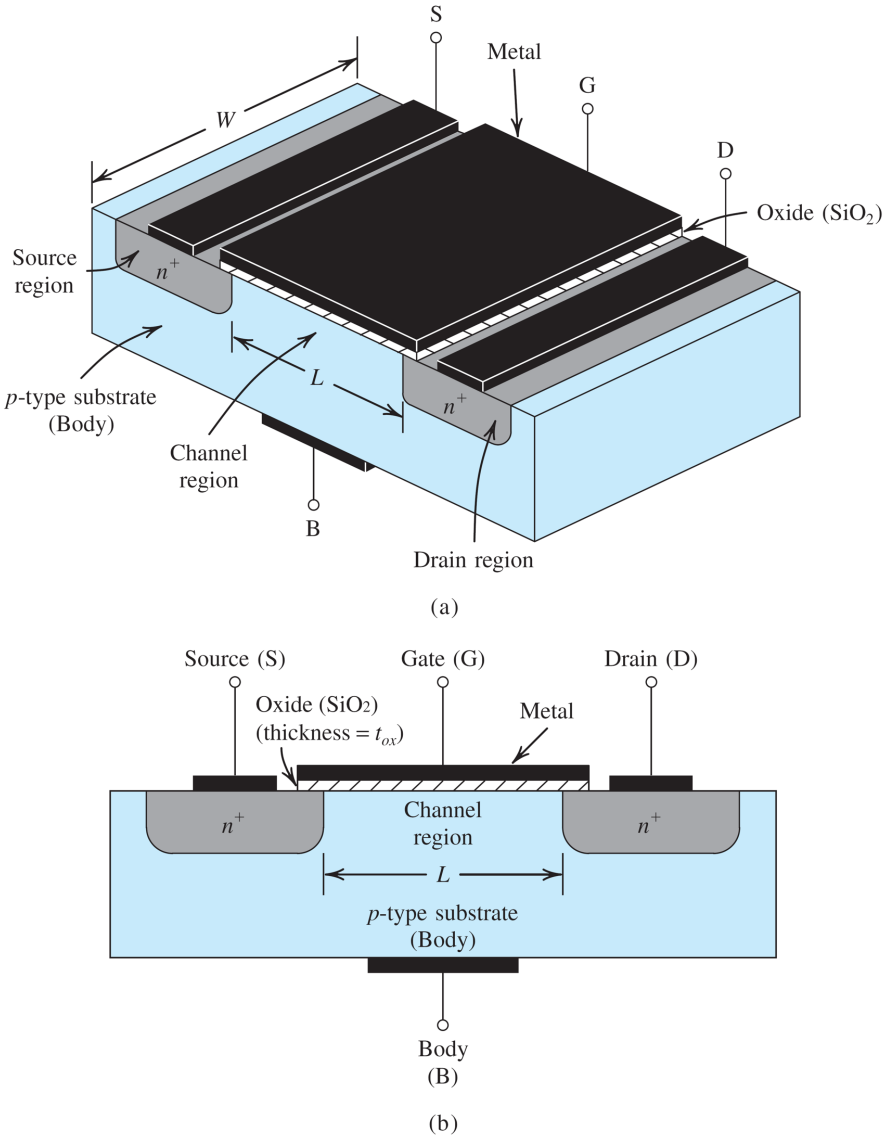


Figure 5.1 Physical structure of the *n*-channel enhancement-type MOSFET: **(a)** perspective view; **(b)** cross section. Typically, $L = 20$ nm to 1 μm, $W = 30$ nm to 100 μm, and the thickness of the oxide layer (t_{ox}) is in the range of 1 to 10 nm.

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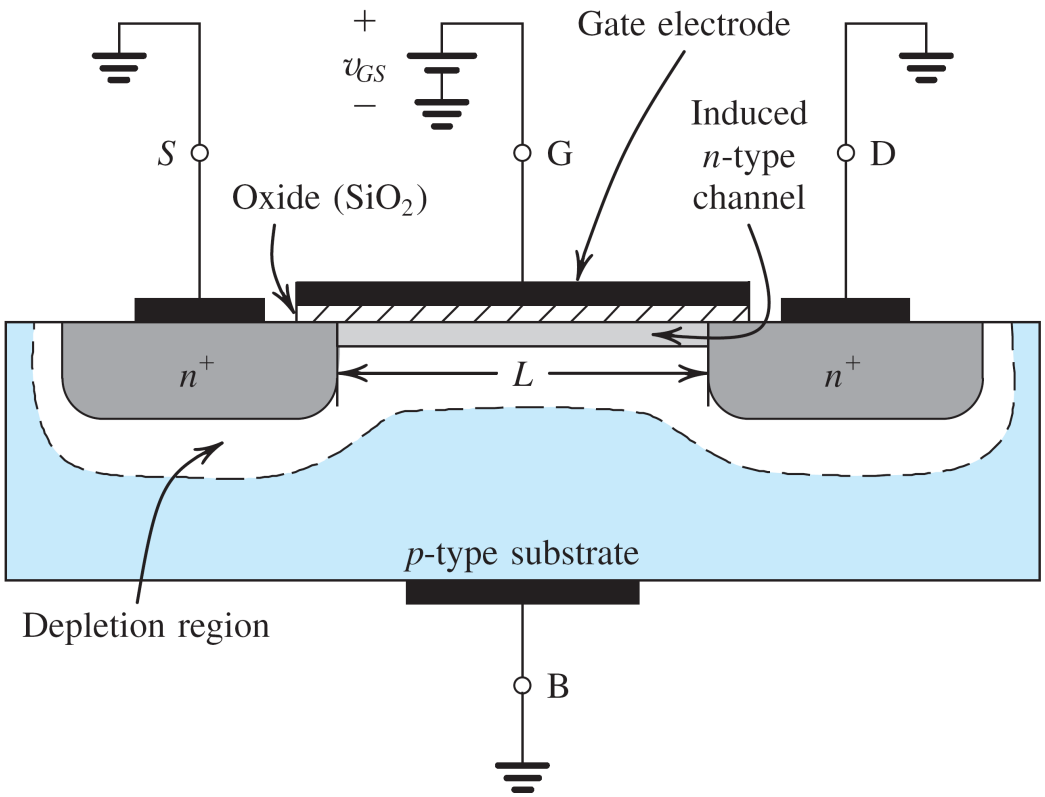


Figure 5.2 The enhancement-type NMOS transistor with a positive voltage applied to the gate. An *n* channel is induced at the top of the substrate beneath the gate.

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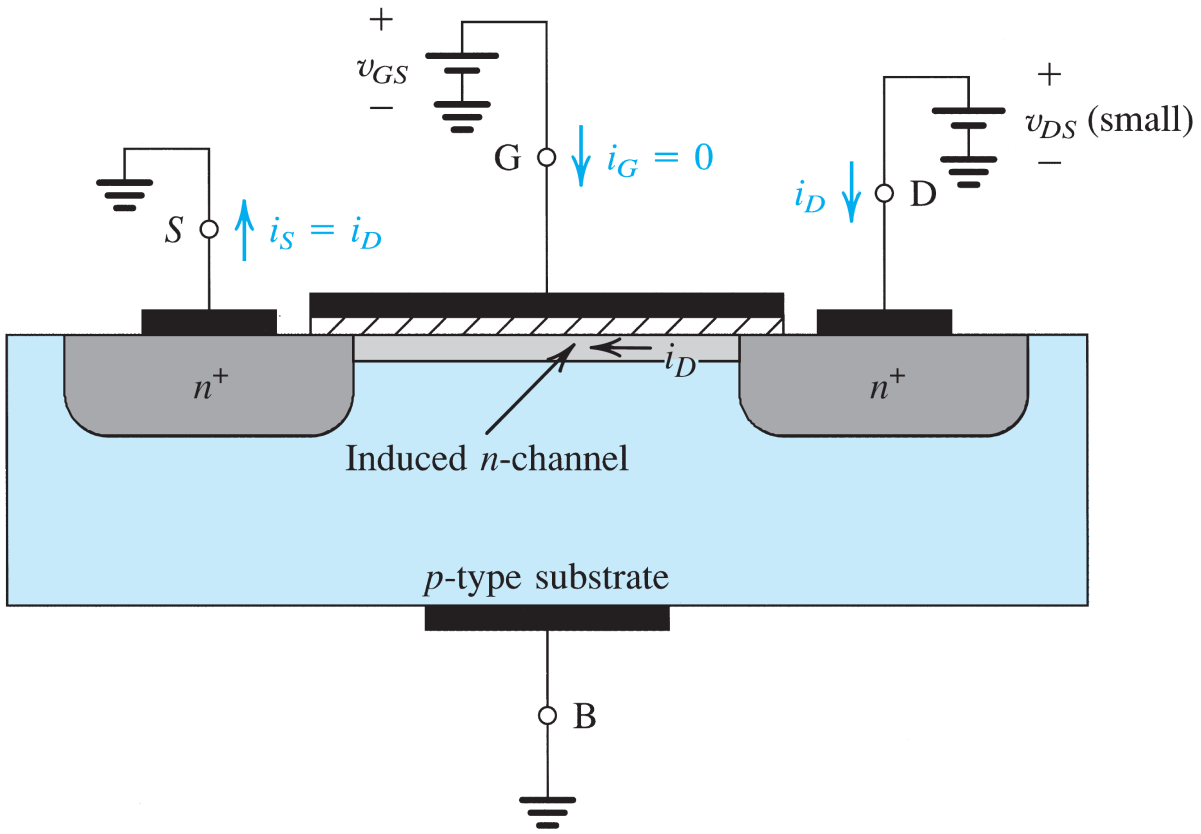


Figure 5.3 An NMOS transistor with $v_{GS} > V_t$ and with a small v_{DS} applied. The device acts as a resistance whose value is determined by v_{GS}. Specifically, the channel conductance is proportional to $v_{GS} - V_t$, and thus i_D is proportional to $(v_{GS} - V_t)v_{DS}$. Note that the depletion region is not shown (for simplicity).

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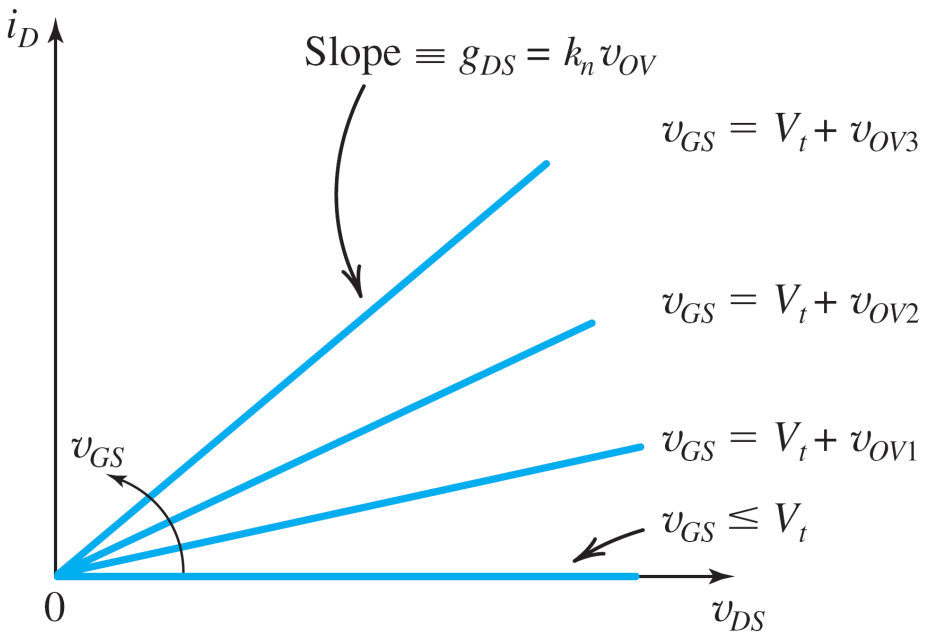


Figure 5.4 The i_D–v_{DS} characteristics of the MOSFET in Fig. 5.3 when the voltage applied between drain and source, v_{DS}, is kept small. The device operates as a linear resistance whose value is controlled by v_{GS}.

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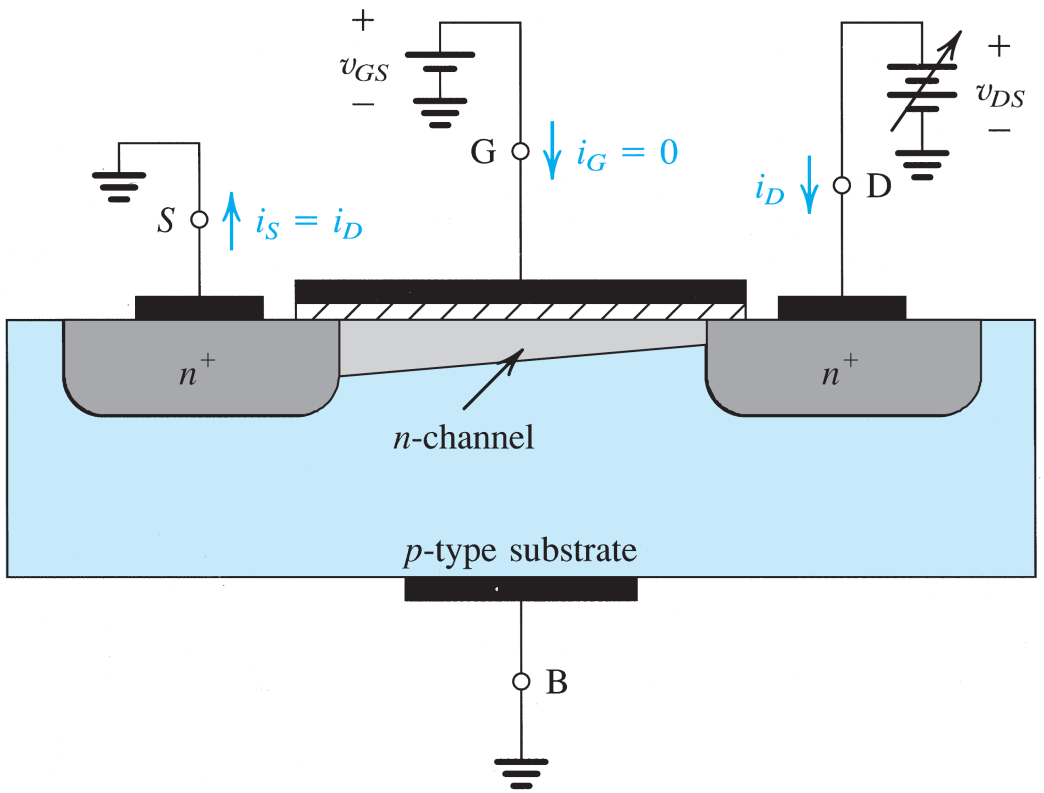


Figure 5.5 Operation of the MOSFET as v_{DS} is increased. The induced channel acquires a tapered shape, and its resistance increases as v_{DS} is increased. Here, v_{GS} is kept constant at a value $> V_t$; $v_{GS} = V_t + v_{OV}$. It is the widening of the depletion region (not shown) as a result of the increased v_{DS} that makes the channel shallower near the drain.

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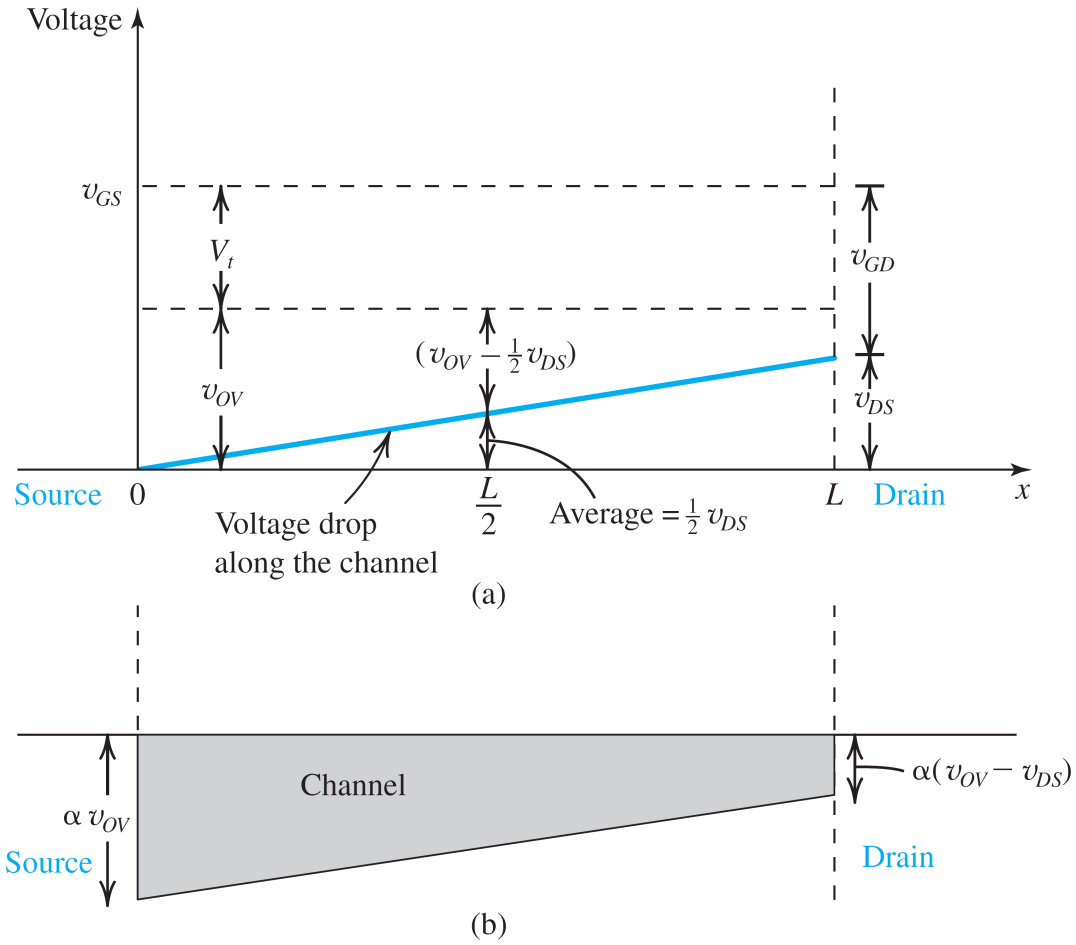


Figure 5.6 **(a)** For a MOSFET with $v_{GS} = V_t + v_{OV}$, applying of v_{DS} causes the voltage drop along the channel to vary linearly, with an average value of $\frac{1}{2}v_{DS}$ at the midpoint. Since $v_{GD} > V_t$, the channel still exists at the drain end. **(b)** The channel shape corresponding to the situation in **(a)**. While the depth of the channel at the source end is still proportional to v_{OV}, that at the drain end is proportional to $(v_{OV} - v_{DS})$.

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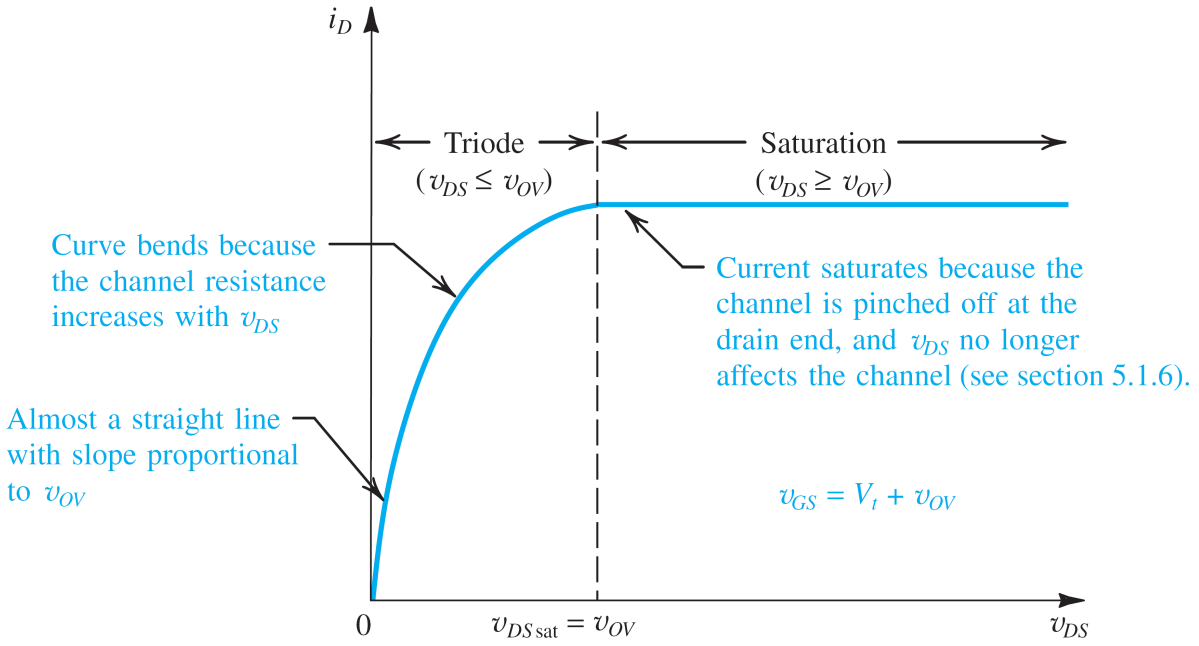


Figure 5.7 The drain current i_D versus the drain-to-source voltage v_{DS} for an enhancement-type NMOS transistor operated with $v_{GS} = V_t + v_{OV}$.

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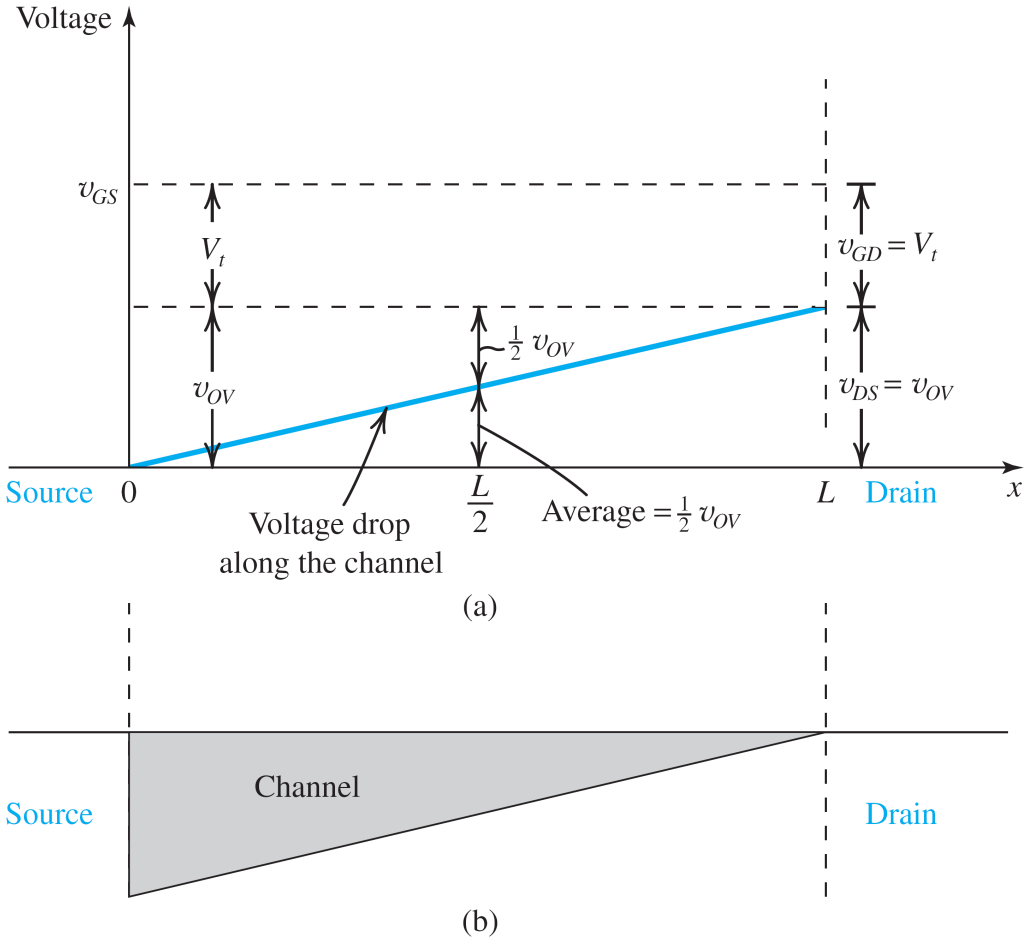


Figure 5.8 Operation of MOSFET with $v_{GS} = V_t + v_{OV}$, as v_{DS} is increased to v_{OV}. At the drain end, v_{GD} decreases to V_t and the channel depth at the drain end reduces to zero (pinch-off). At this point, the MOSFET enters the saturation mode of operation. Further increasing v_{DS} (beyond $v_{DSsat} = v_{OV}$) has no effect on the channel shape and i_D remains constant.

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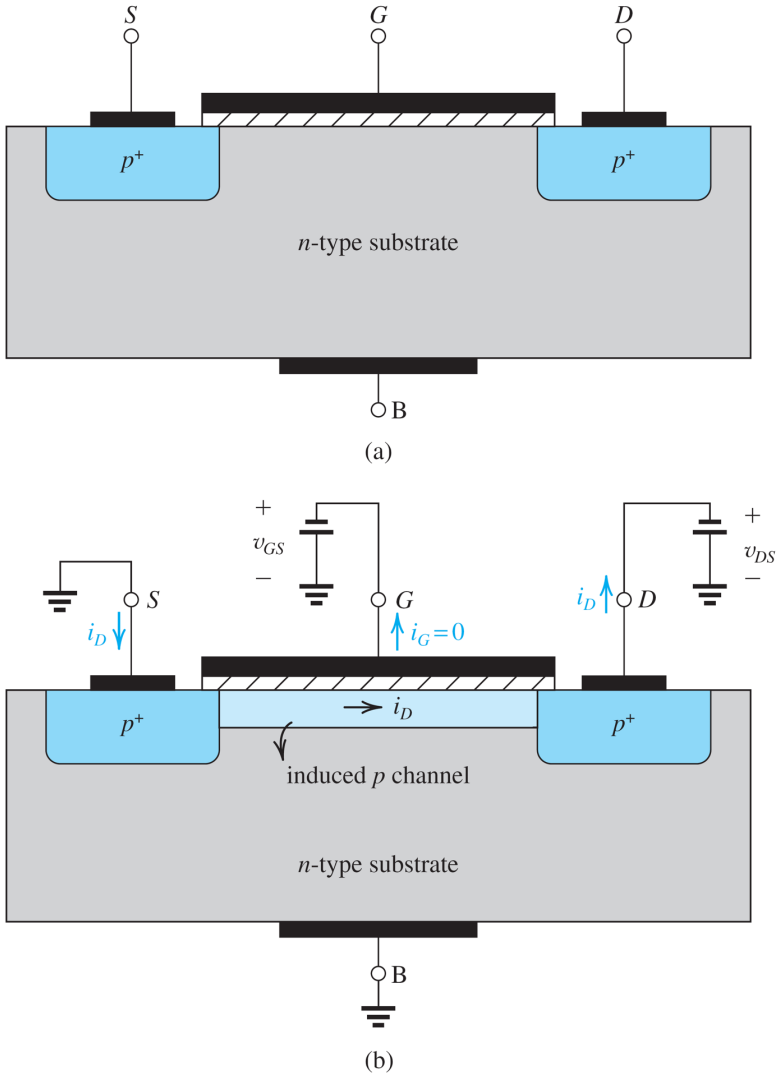


Figure 5.9 **(a)** Physical structure of the PMOS transistor. Note that it is similar to the NMOS transistor shown in Fig. 5.1(b) except that all semiconductor regions are reversed in polarity. **(b)** A negative voltage v_{GS} of magnitude greater than $|V_{tp}|$ induces a p channel, and a negative v_{DS} causes a current i_D to flow from source to drain.

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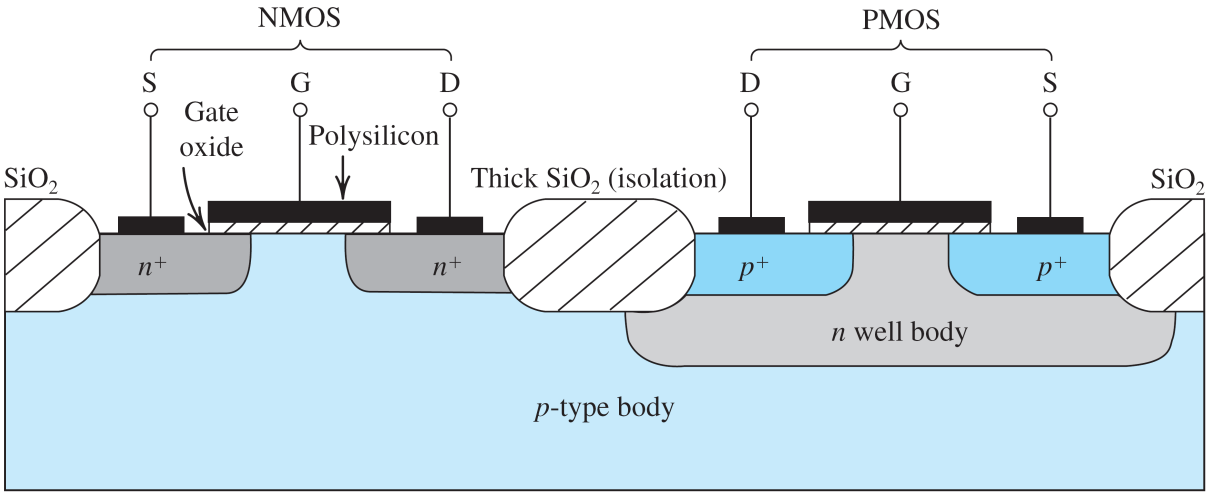


Figure 5.10 Cross section of a CMOS integrated circuit. Note that the PMOS transistor is formed in a separate *n*-type region, known as an *n* well, which functions as the body for the PMOS transistor. Not shown are the connections made to the *p*-type body and to the *n* well body.

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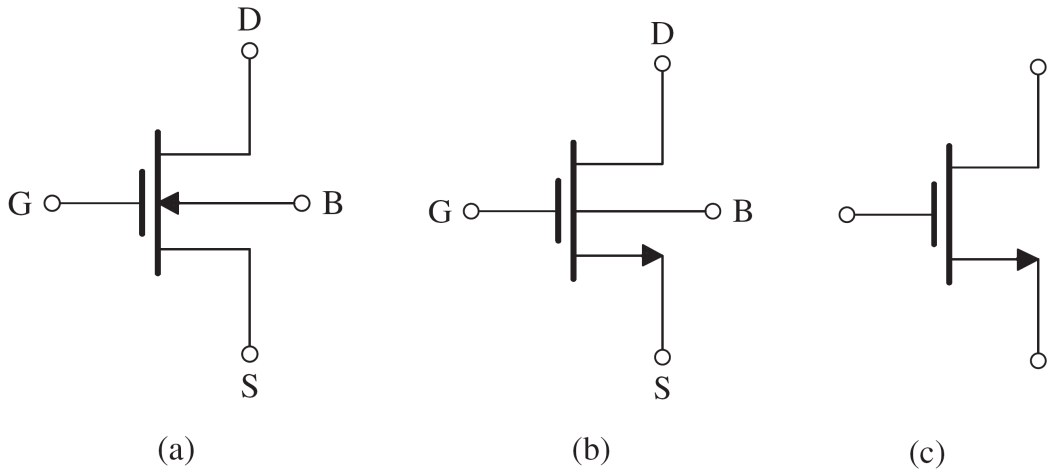
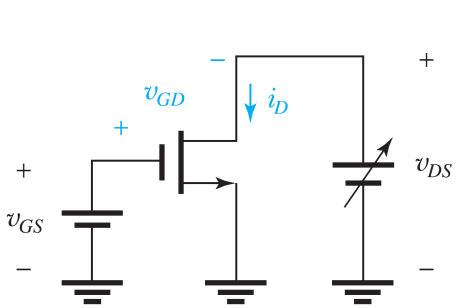
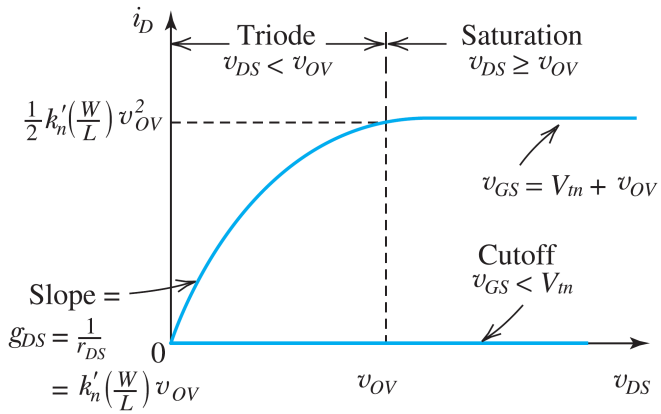


Figure 5.11 **(a)** Circuit symbol for the *n*-channel enhancement-type MOSFET. **(b)** Modified circuit symbol with an arrowhead on the source terminal to distinguish it from the drain and to indicate device polarity (i.e., *n* channel). **(c)** Simplified circuit symbol to be used when the source is connected to the body or when the effect of the body on device operation is unimportant.

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Table 5.1 Regions of Operation of the Enhancement NMOS Transistor





- $v_{GS} < V_{tn}$: no channel; transistor in cutoff; $i_D = 0$
- $v_{GS} = V_{tn} + v_{OV}$: a channel is induced; transistor operates in the triode region or the saturation region depending on whether the channel is continuous or pinched off at the drain end;

Triode Region

Continuous channel, obtained by:

$$v_{GD} > V_{tn}$$

or equivalently:

$$v_{DS} < v_{OV}$$

Then,

$$i_D = k_n'\left(\frac{W}{L}\right)\left[(v_{GS} - V_{tn})v_{DS} - \frac{1}{2}v_{DS}^2\right]$$

or equivalently,

$$i_D = k_n'\left(\frac{W}{L}\right)\left(v_{OV} - \frac{1}{2}v_{DS}\right)v_{DS}$$

Saturation Region

Pinched-off channel, obtained by:

$$v_{GD} \leq V_{tn}$$

or equivalently:

$$v_{DS} \geq v_{OV}$$

Then

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - V_{tn})^2$$

or equivalently,

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)v_{OV}^2$$

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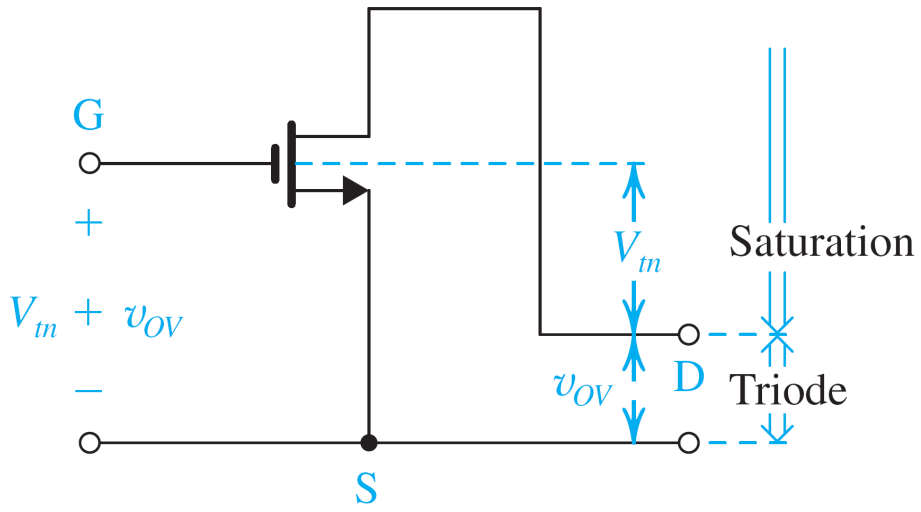


Figure 5.12 The relative levels of the terminal voltages of the enhancement NMOS transistor for operation in the triode region and in the saturation region.

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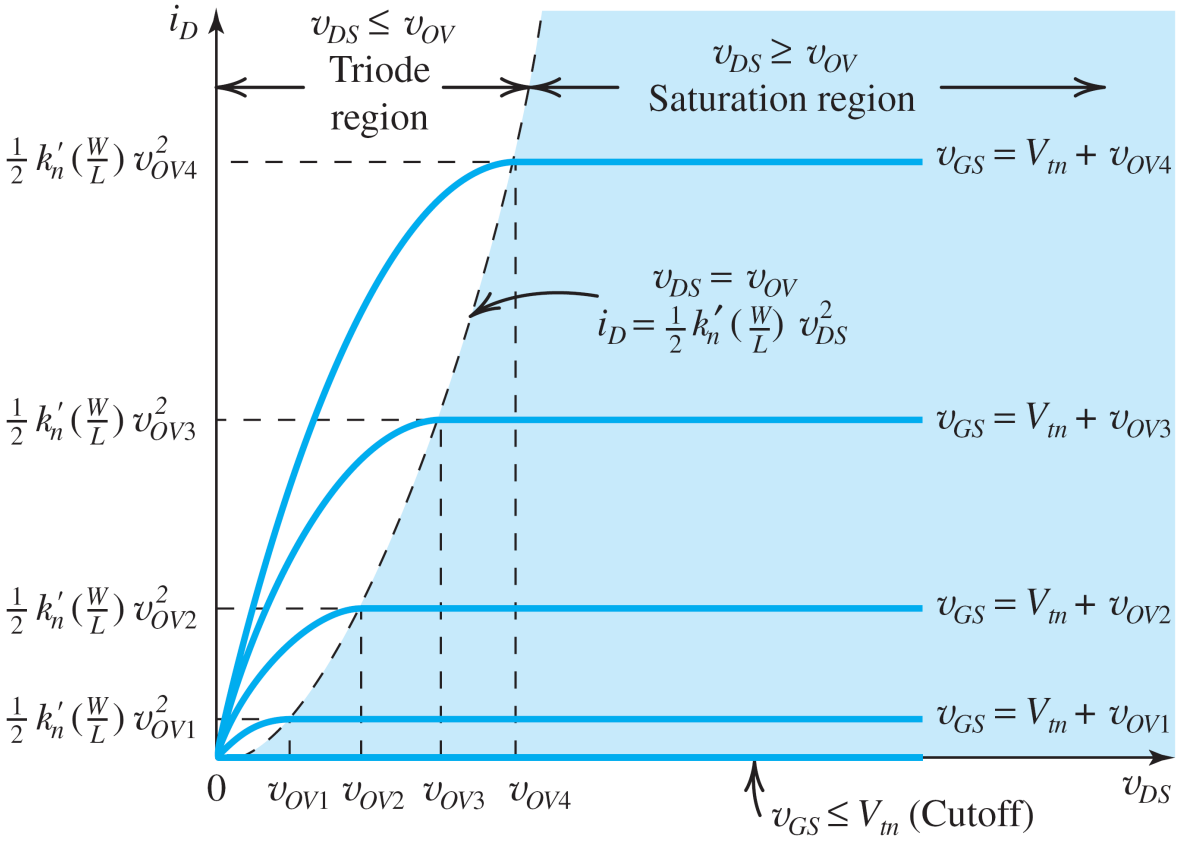


Figure 5.13 The $i_D - v_{DS}$ characteristics for an enhancement-type NMOS transistor.

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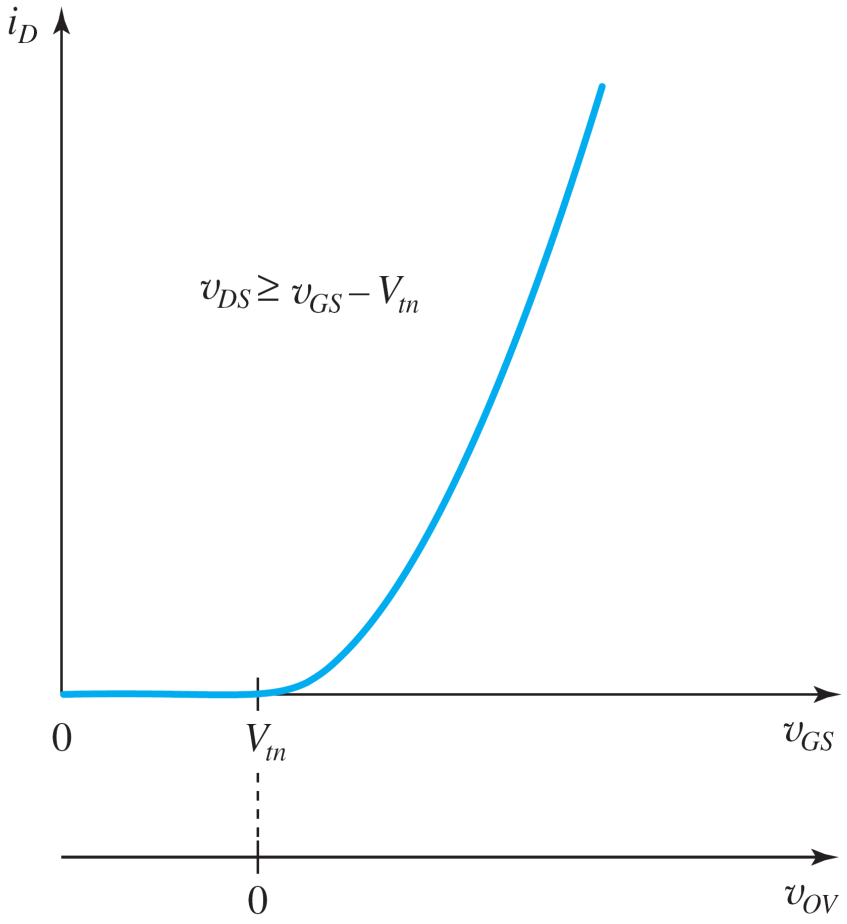


Figure 5.14 The i_D–v_{GS} characteristic of an NMOS transistor operating in the saturation region. The i_D–v_{OV} characteristic can be obtained by simply relabeling the horizontal axis, that is, shifting the origin to the point $v_{GS} = V_{tn}$.

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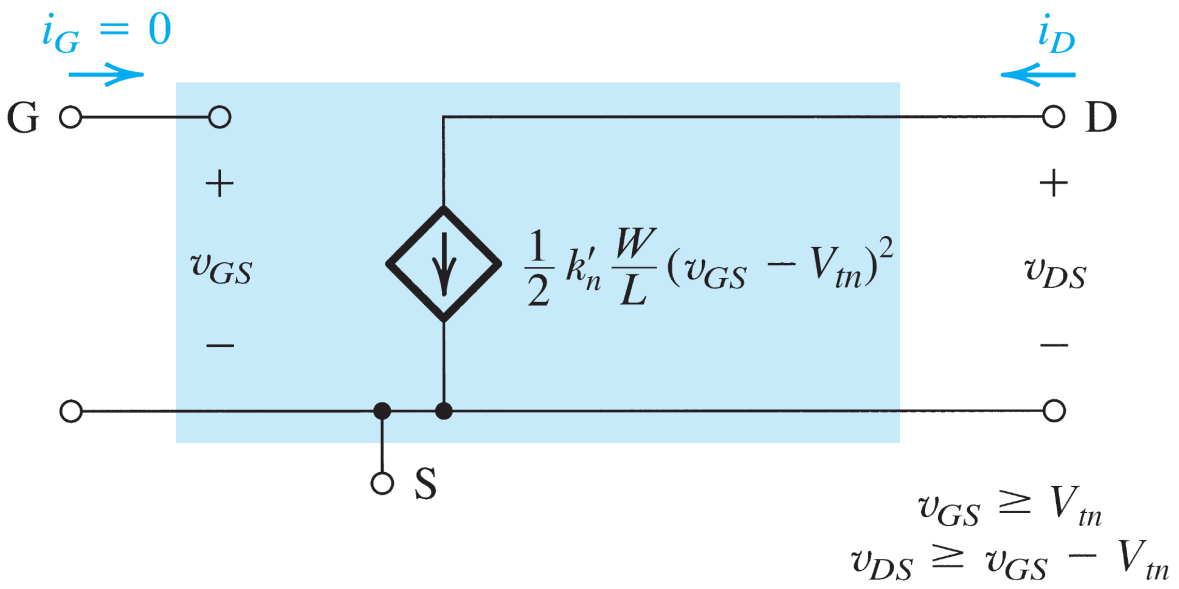


Figure 5.15 Large-signal, equivalent-circuit model of an *n*-channel MOSFET operating in the saturation region.

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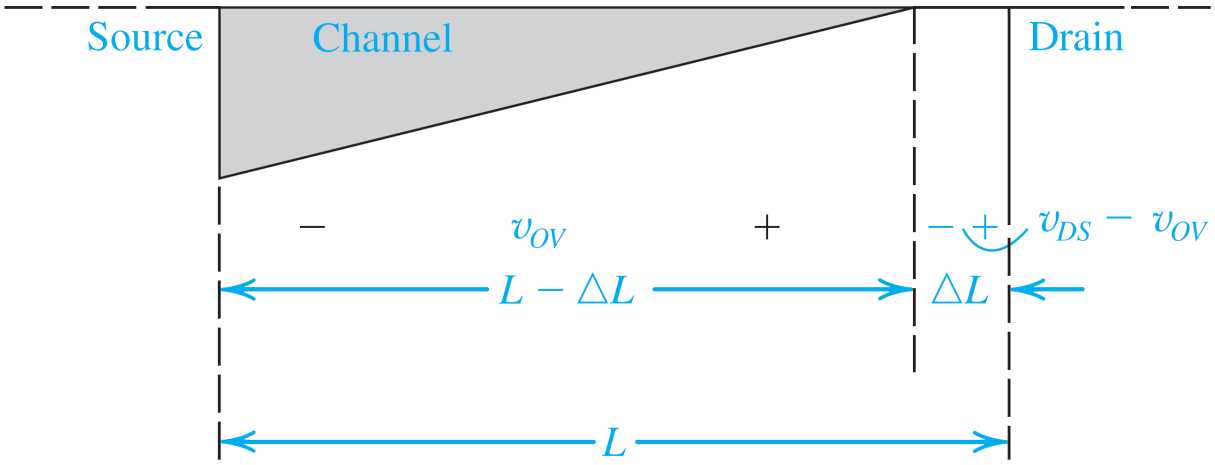


Figure 5.16 Increasing v_{DS} beyond v_{DSsat} (which is equal to v_{OV}) causes the channel pinch-off point to move slightly away from the drain, thus reducing the effective channel length (by ΔL).

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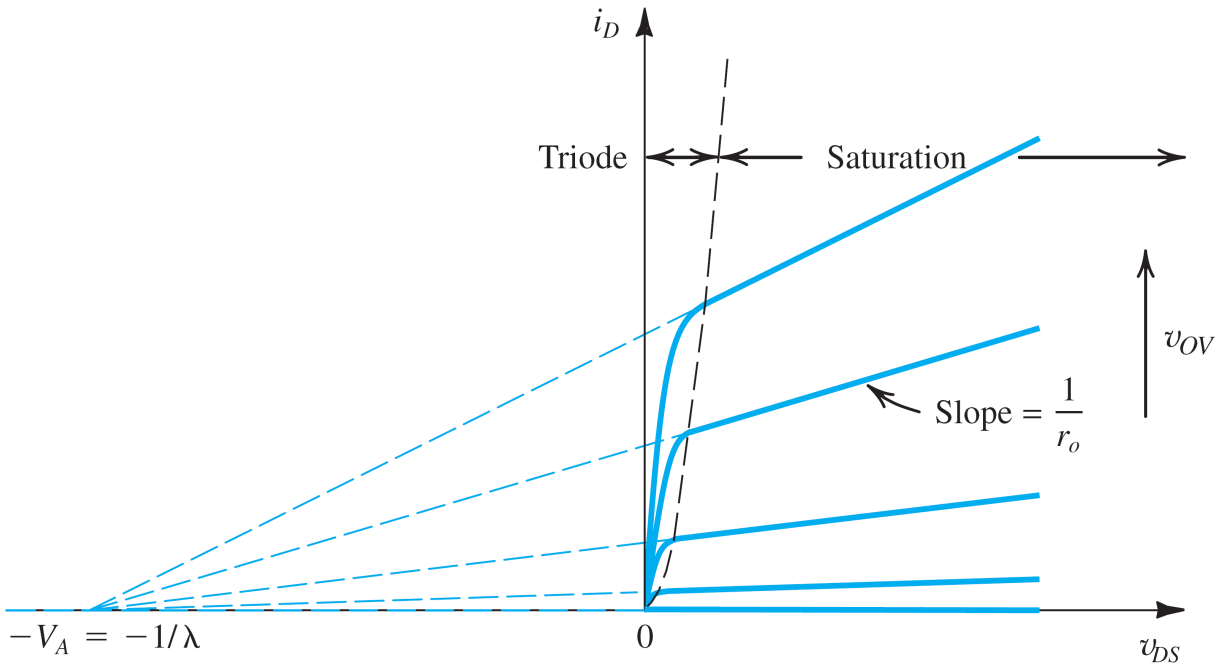


Figure 5.17 Effect of v_{DS} on i_D in the saturation region. The MOSFET parameter V_A depends on the process technology and, for a given process, is proportional to the channel length L.

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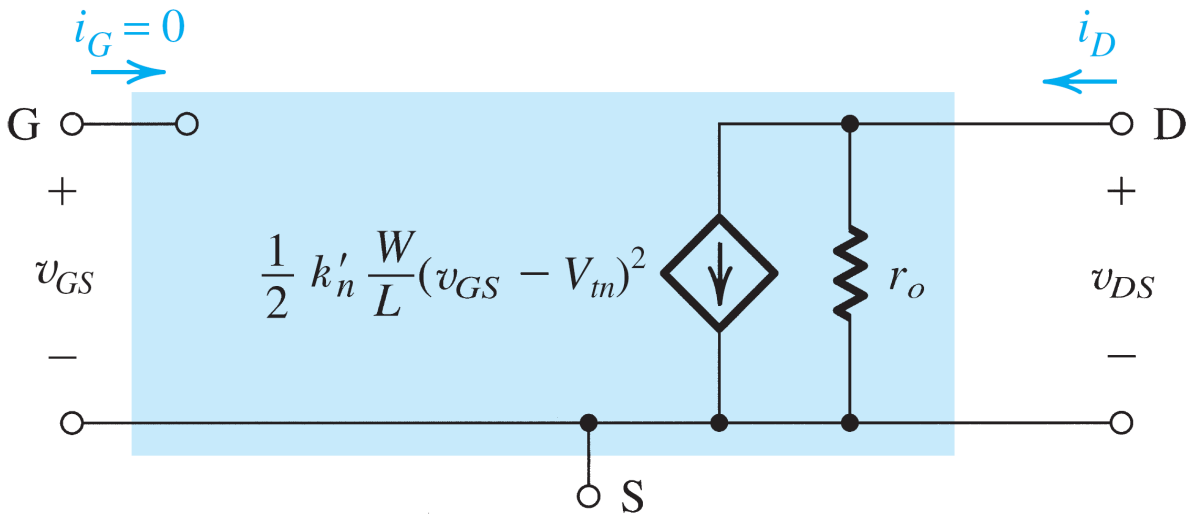


Figure 5.18 Large-signal, equivalent-circuit model of the *n*-channel MOSFET in saturation, incorporating the output resistance r_o. The output resistance models the linear dependence of i_D on v_{DS} and is given by Eq. (5.27).

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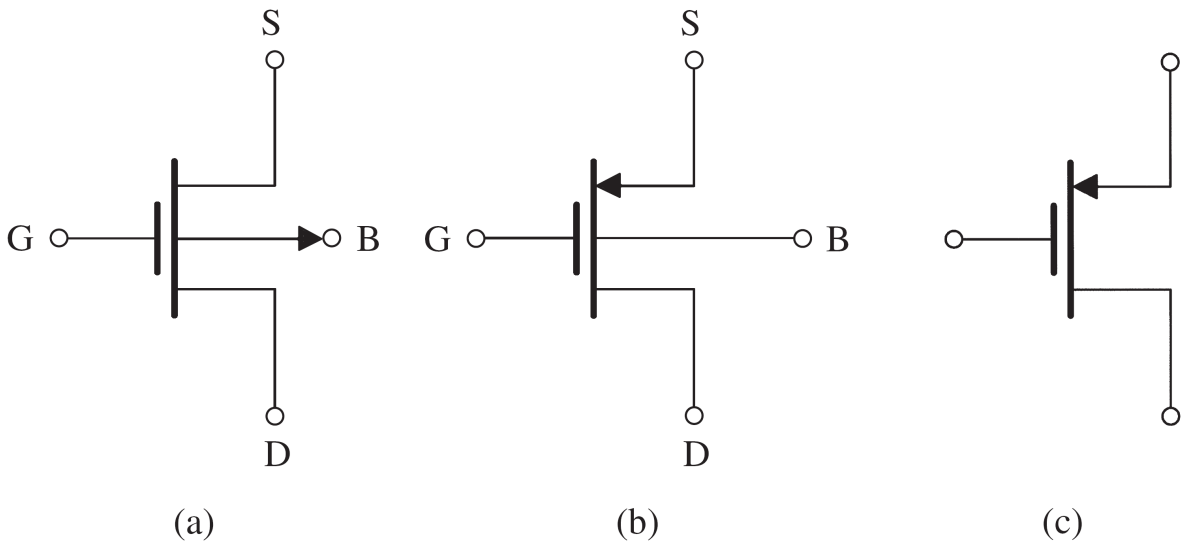
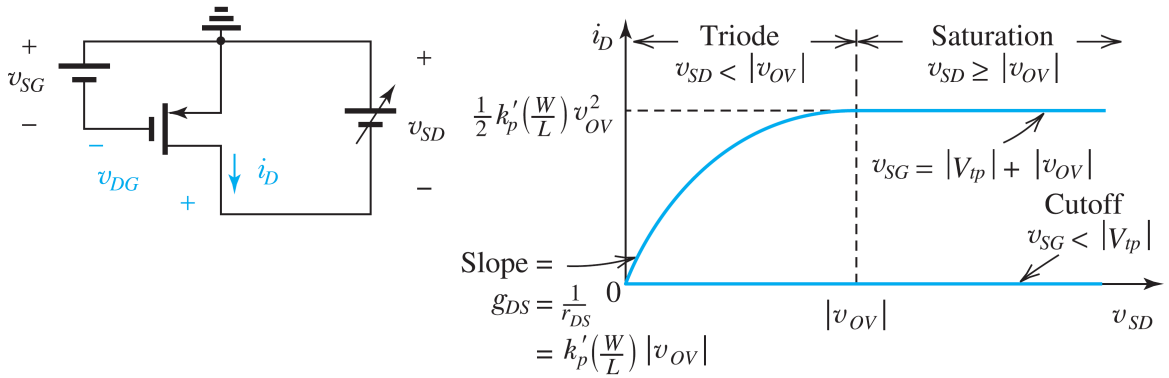


Figure 5.19 **(a)** Circuit symbol for the *p*-channel enhancement-type MOSFET. **(b)** Modified symbol with an arrowhead on the source lead. **(c)** Simplified symbol for cases where the source is connected to the body.

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Table 5.2 Regions of Operation of the Enhancement PMOS Transistor



- $v_{SG} < |V_{tp}|$: no channel; transistor in cutoff; $i_D = 0$
- $v_{SG} = |V_{tp}| + |v_{OV}|$: a channel is induced; transistor operates in the triode region or in the saturation region depending on whether the channel is continuous or pinched off at the drain end;

Triode Region

Continuous channel, obtained by:

$$v_{DG} > |V_{tp}|$$

or equivalently

$$v_{SD} < |v_{OV}|$$

Then

$$i_D = k_p'\left(\frac{W}{L}\right)\left[(v_{SG} - |V_{tp}|)v_{SD} - \frac{1}{2}v_{SD}^2\right]$$

or equivalently

$$i_D = k_p'\left(\frac{W}{L}\right)\left(|v_{OV}| - \frac{1}{2}v_{SD}\right)v_{SD}$$

Saturation Region

Pinched-off channel, obtained by:

$$v_{DG} \leq |V_{tp}|$$

or equivalently

$$v_{SD} \geq |v_{OV}|$$

Then

$$i_D = \frac{1}{2}k_p'\left(\frac{W}{L}\right)(v_{SG} - |V_{tp}|)^2$$

or equivalently

$$i_D = \frac{1}{2}k_p'\left(\frac{W}{L}\right)v_{OV}^2$$

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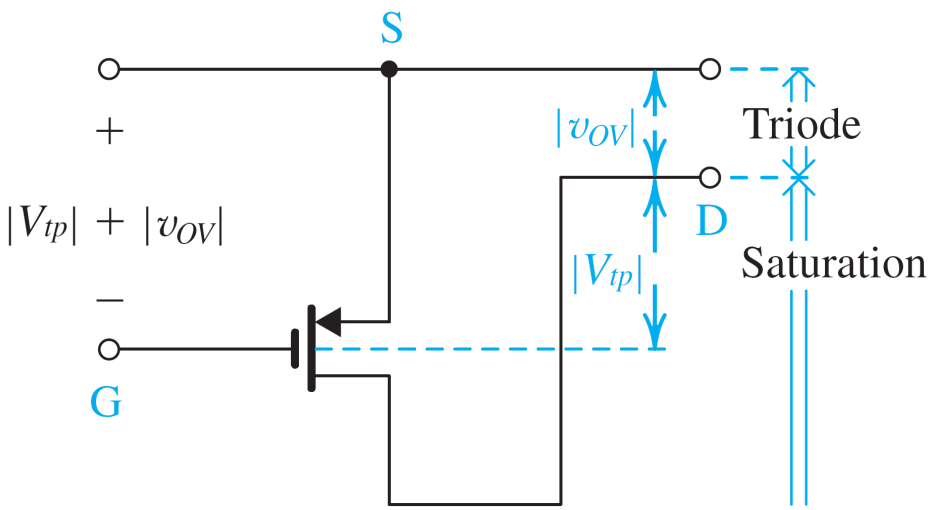


Figure 5.20 The relative levels of the terminal voltages of the enhancement-type PMOS transistor for operation in the triode region and in the saturation region.

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